



Transcribe Audio Files with AI



Arpita Chowdhury

The screenshot displays the 'Transcription preview' section of the Nextwork.org interface. At the top right of this section is a 'Download' button with a dropdown arrow. Below the button is a text prompt: 'Select download to save a local copy of the transcription.' A horizontal tab bar contains four options: 'Text', 'Audio identification' (which is selected and highlighted in blue), 'Subtitles', and 'Toxicity detection - new'. The main content area shows a transcript with three entries:

- Speaker 0:** Indicates that Docker couldn't pull a Docker image from Amazon ECR, OK. The specific error is a 403 forbidden response which usually points to an issue with permissions or access rights. OK, that's what we thought, which is cool. Uh-huh. Since repositories with ECR are private by default, both you and Player A have set up private repositories that no-one else can access until you give someone else specific permissions.
- Speaker 1:** ***, so what we've set up are private repositories, my friends. Do you see even in the left-hand navigation panel for Maximus is images that highlight a thing it's under the heading private registry. ***, so what's really happening here is it's so private that no one else, *** no one else can have access until we give it to them specifically, and that's the reason why we can just openly paste the links to our registries in the chat because *** we know that even if you have access to that link, you can't *** get access to the registry itself until we give you specific access.
- Speaker 0:** Cool.



Introducing Today's Project!

In this project, I will demonstrate how to use Amazon Transcribe to convert audio and video files into accurate, searchable text using AWS services. I'm doing this project to learn how AI-powered speech-to-text tools can be integrated into real applications, including storing media in S3, improving transcription accuracy with custom vocabularies and filters, and transcribing live speech—skills that are especially valuable for Data Analyst, AI Developer, and Cloud Architect roles.

Tools and concepts

Services I used were Amazon Transcribe and Amazon S3 to store audio/video files, generate batch and real-time transcriptions, apply custom vocabularies and filters, and produce subtitles. Key concepts I learnt include speech-to-text transcription, baseline vs enhanced transcription, custom vocabularies, vocabulary filtering, subtitle generation (SRT), speaker partitioning (diarization), and real-time streaming transcription, along with understanding how these features improve accuracy, accessibility, and usability in real-world applications.

Project reflection

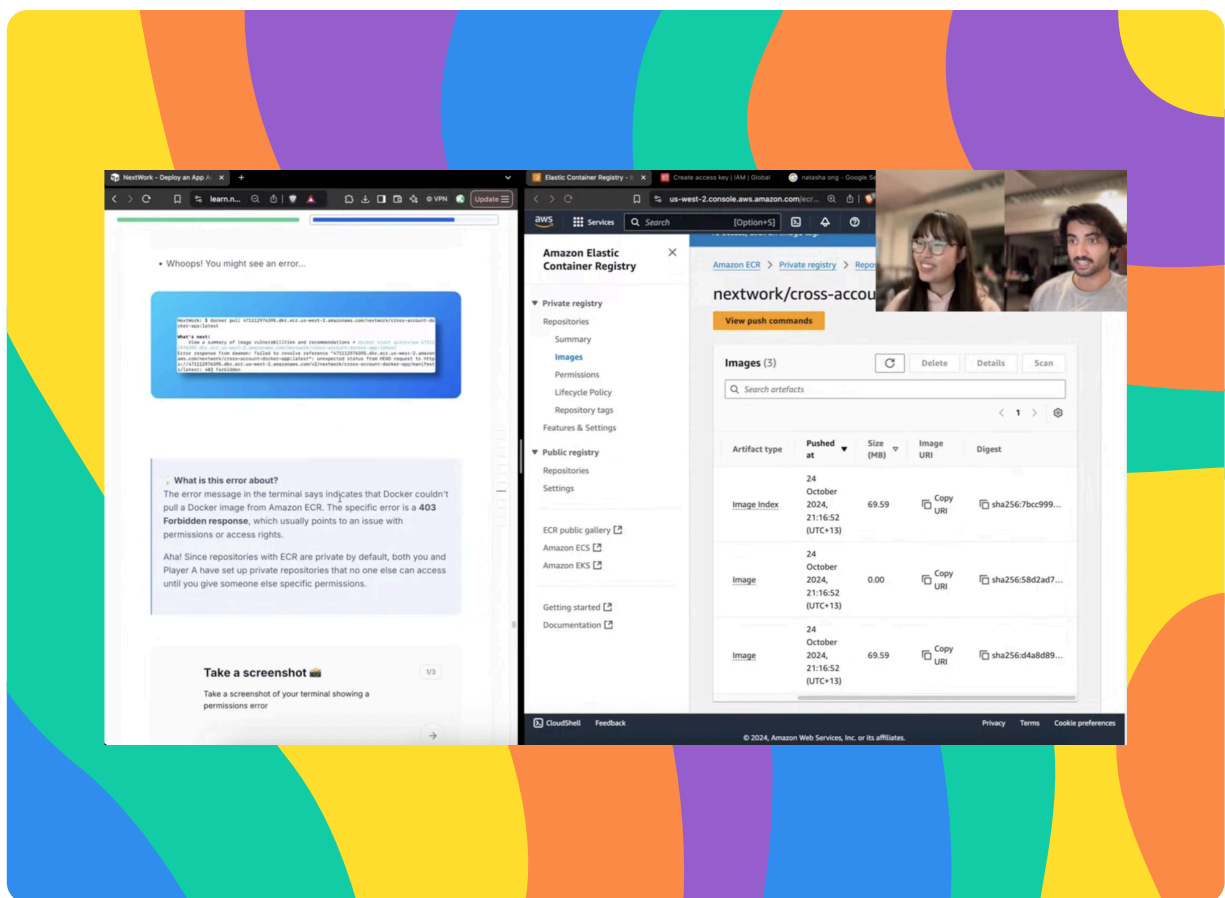
This project took me approximately 2 hours. The most challenging part was fine-tuning the transcription using custom vocabularies and vocabulary filters to improve accuracy while removing unwanted filler words. It was most rewarding to see a clear improvement between the baseline and enhanced transcriptions and experience real-time transcription generating text instantly as I spoke.

I chose to do this project today because I wanted hands-on experience using AI-powered speech-to-text tools to turn audio and video into usable, searchable data, which is a valuable skill for data, AI, and cloud roles. Something that would make learning with NextWork even better is including more real-world datasets and industry use cases, along with side-by-side comparisons of baseline and enhanced outputs to clearly show the impact of each feature.



S3 and Transcribe

To set up for this project, I'm using an S3 bucket to store my video file securely and make it accessible to Amazon Transcribe, because Transcribe requires media files to be hosted in S3 before they can be processed. The file I'm transcribing is a video containing spoken audio that I want to convert into text.





Run A Transcription Job

The steps to run a transcription job include selecting the media file stored in Amazon S3, choosing the appropriate transcription model and language, configuring basic job settings, and starting the transcription job to generate the text output. Overall, this process took me just a few minutes, as Amazon Transcribe automatically handles the processing once the job is submitted.

Amazon Transcribe uses model types to improve transcription accuracy by tailoring its speech-to-text processing to different kinds of audio content. Since conversations, lectures, customer service calls, and media recordings all have unique speaking patterns and vocabulary, model types help Transcribe select the most suitable AI model for each scenario. Common use cases include general speech such as meetings and lectures, customer service and call-center conversations, media content like podcasts and interviews, and specialized domains such as medical discussions where technical terminology is used.

You can customise a transcription further with subtitling, which adds time-aligned text captions (like SRT files) so spoken words appear as readable subtitles in videos, and speaker partitioning, which helps to identify and separate different speakers in the audio so you can clearly see who said what in conversations.



Vocabulary filtering [Info](#)
Vocabulary filtering can remove, mask or tag specified words in the final transcript.

Filter selection
The vocabulary filters shown here are based on your language settings. You can choose up to one vocabulary filter per language. You can also create a new vocabulary filter. [↗](#)

filter-words-filter ▼

Vocabulary filtering method
Use a vocabulary filter to filter vocabularies from your transcript. For example, in the sentence "The quick brown fox jumps over the lazy dog," you remove the word "lazy" using the following options.

☒ **Mask vocabulary**
Example: The quick brown fox jumps over the *** dog.

☐ **Remove vocabulary**
Example: The quick brown fox jumps over the dog.

☐ **Tag vocabulary**
Example: The quick brown fox jumps over the lazy_dog.

Customization

☒ **Custom vocabulary** [Info](#)
A custom vocabulary improves the accuracy of recognizing words and phrases specific to your use case.

Vocabulary selection
The vocabularies shown here are based on your language settings. You can choose up to one vocabulary per language. You can also create a new vocabulary. [↗](#)

nextwork-project-vocab ▼

Cancel Previous **Create job**



Baseline Transcript Review

To start using Amazon Transcribe, I first ran a baseline transcription job, which means creating a standard transcription without any custom vocabularies or filters applied. This is because it gives a reference point to evaluate the transcription's default accuracy and helps identify what improvements are needed before adding custom enhancements.

While reviewing the baseline transcript, I noticed filler words (such as “uh-huh” and “yeah”), repeated phrases, run-on sentences, and some awkward punctuation and formatting, which made the text less clean and harder to read. These happened because the baseline transcription captures spoken language very literally without applying custom vocabularies, filters, or post-processing rules, so it includes natural speech patterns like repetition, informal phrasing, and pauses that are common in real conversations.



Indicates that Docker couldn't pull a Docker image from Amazon ECR, OK. The specific error is a 403 forbidden response which usually points to an issue with permissions or access rights. OK, that's what we thought, which is cool. Uh-huh. Since repositories with ECR are private by default, both you and player A have set up private repositories that no-one else can access until you give someone else specific permissions. Yeah, so what we've set up are private repositories, my friends. Do you see even in the left-hand navigation panel for Maximus is images that highlight a thing it's under the heading private registry. Um, so what's really happening here is it's so private that no one else, literally no one else can have access until we give it to them specifically, and that's the reason why we can just openly paste the links to our registries in the chat because actually we know that even if you have access to that link, you can't actually get access to the registry itself until we give you specific access. Cool. So let's take a screenshot. Let's take a screenshot. So we're gonna go back to our terminal. And just take a screenshot of this error right here. Nice. And I'm gonna paste that into a box here and then move on to the next question.



Custom Vocabulary

I can resolve transcription inaccuracies using a custom vocabulary, which is a list of words or phrases that I define to tell Amazon Transcribe how specific terms should be recognized and displayed in the transcript. A custom vocabulary improves transcription accuracy by teaching Transcribe the correct spelling, formatting, and capitalization of domain-specific terms, technical language, or commonly misinterpreted words.

To create an item in a custom vocabulary, you need to define two values: Phrase – the word or term you want Amazon Transcribe to recognize. Display as – how that word should appear in the final transcription (for example, with correct capitalization or formatting).

My custom vocabulary defines how specific words and phrases from the audio should be transcribed correctly, including capitalizing “player” as “Player,” correcting misspelled terms like “repositories,” and displaying spoken phrases like “four-oh-three-forbidden” as “403 Forbidden” in the final transcript.



View and edit vocabulary - new (3) [Info](#)

Reset vocabulary

Delete

Download latest ▼

🔍 Filter Phrase, SoundsLike, IPA or DisplayAs

Show all ▼

< 1 >

☐	Phrase 🔗 ▼	SoundsLike - optional 🔗 ▼	IPA - optional 🔗 ▼	DisplayAs - optional 🔗 ▼
☐	player	-	-	Player
☐	repositories	-	-	repositories
☐	four-or-three-forbidden	-	-	403 Forbidden 🔗

Add row



Vocabulary Filters

Another feature in Amazon Transcribe is vocabulary filtering, which allows you to automatically remove, mask, or flag specific unwanted words or phrases from a transcription. It's different from custom vocabularies because custom vocabularies improve how words are recognized and displayed, while vocabulary filtering is used to control or censor words after they've been transcribed.

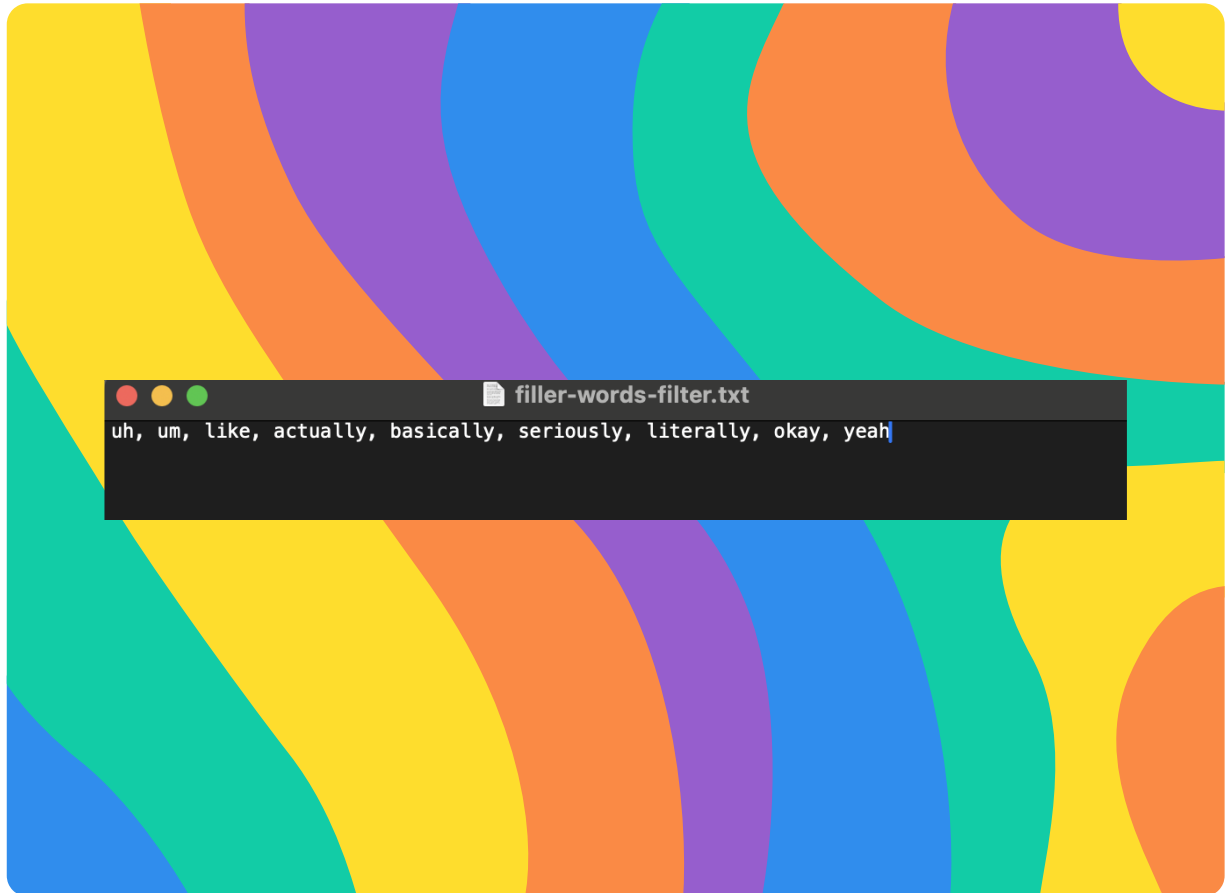
My vocabulary filter removes common filler words such as "uh," "um," "like," "actually," "basically," "seriously," "literally," "okay," and "yeah" from the transcription. To set up this filter, I first created a text file listing these filler words and then used it to create a vocabulary filter in Amazon Transcribe.



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Enhanced Transcription

I ran a new transcription with my custom vocabulary and filtering settings

The enhanced transcription is better than the baseline because it is cleaner, more readable, and more accurate, with filler words removed, technical terms correctly formatted (such as “403 Forbidden”), proper capitalization applied, and clearer speaker separation, making the transcript easier to understand and more useful for analysis or subtitles.



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Real Time Transcription

For my project extension, I experimented with real-time transcription, which converts speech into text instantly as I speak, allowing me to see the transcription generated live. This demonstrates how Amazon Transcribe can be used for live captions, voice-controlled applications, and accessibility features in real-world systems.

Even during real-time transcription, I could use features like custom vocabularies and vocabulary filtering to improve accuracy and clean up the live transcript as speech was being processed. Overall, compared to a transcription job, real-time transcription was faster and more interactive, providing instant results as I spoke, whereas a transcription job is better suited for processing and refining pre-recorded audio or video files.



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Real-time transcription Info

See how Amazon Transcribe creates a text copy of speech in real time. Choose **Start streaming** and talk.

Transcription

[Download full transcript](#)

 **Start streaming**

Transcription output

Current language: English, US

Hi, my name is Orpita.
I am Player a in a project on Docker and container.

there was a 403 forbidden response.



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